

Coupling-gated cortical coherence: what a field theory of conscious unity must assume, and what it predicts

Sebastian Bobadilla-Suarez
`sebastian.bobadilla.s@gmail.com`

6 September 2026

Abstract

A field account of cortical coordination holds that spatially distributed neural processes are coupled in part through the endogenous extracellular electromagnetic field. We ask what such an account must assume to reach a globally consistent, self-referential state, and what it predicts before it gets there. Its physical core is a noisy mean-field phase model whose coupling drifts through a synchronization threshold. If recovery of coherence after sleep is gated by that crossing, it is not free to take an arbitrary time course. For as long as the phase distribution tracks the stationary branch, it must show a square-root onset and a von Mises concentration–coherence relation with no fitted shape parameter. The tracking condition is part of the prediction, and the simulations measure its cost: ramp speed sets the apparent onset exponent, so an exponent fitted to a driven recovery trace reports the drive as much as the mechanism. A coupling plasticity descending a squared-drift objective preserves coherence while moving away from the environmental structure it should learn, below a shuffled-label comparison: dissipation minimization does not imply representational learning. A Lean 4 formalization composes finite capacity, Landauer’s

bound, the noisy Kuramoto transition, sheaf gluing and Banach’s theorem into one conditional theorem, isolating the eight hypotheses it consumes; a common toy model satisfies all eight. We defend as a philosophical commitment, not a theorem, the identification of the glued section with the unity of a conscious episode and of its reflexive fixed point with a minimal self. Testing the account requires independent calibration of effective cortical coupling and an awakening experiment separating slow coupling change from alternative recovery mechanisms.

1 Introduction

Neural activity changes across sleep, awakening and anesthesia, but a change in synchrony does not by itself identify the physical interaction responsible for it. A field account makes a stronger proposal: spatially distributed neural processes are coordinated partly through the endogenous extracellular electromagnetic field, and changes in the medium can change that coordination (McFadden, 2002, 2020; Pockett, 2012). Ephaptic effects provide a possible interaction mechanism (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Whether that mechanism is sufficient to support the collective state such accounts require remains an empirical question.

Such an account has three requirements, and they order what follows. Coupling must be strong enough to sustain coherent collective activity; the local descriptions that activity carries must be compatible enough to glue into a single one; and that glued state must support an accurate model of itself. The first requirement is physical and testable with current methods, and most of this article is about testing it. The second and third are where

17 the mathematics is, and where the assumptions accumulate.

18 The first requirement admits a definite claim. If cortical coupling is slowly
19 gated—if recovery of coherence after sleep is driven by a coupling parameter
20 drifting through a synchronization threshold—then the recovery is not free
21 to take an arbitrary time course. For a recovery slow enough that the phase
22 distribution tracks the stationary branch, the theory fixes a square-root onset
23 and a one-parameter relation between phase concentration and coherence,
24 with no fitted shape parameter. That is a constraint an experiment can
25 meet or violate. How slow is slow enough is itself part of the claim, and
26 our simulations bound it from the other side, by showing which parts of the
27 constraint survive finite populations and finite ramp speeds and which do
28 not.

29 The contribution has four parts. The first is the coupling-gated recovery
30 hypothesis itself: a stationary coherent branch with square-root onset, a von
31 Mises concentration–coherence relation, and an experimental protocol with
32 failure conditions declared in advance (Sections 2 and 3). The second is a
33 constraint on how such data may be analyzed. The apparent onset expo-
34 nent measured from a finite ramp is a function of ramp speed across the
35 sweep of Section 4, so an exponent fitted to a recovery trace measures the
36 ramp at least as much as the mechanism—and that holds for any analysis
37 reading a critical exponent off a driven transition, not only for ours. The
38 third is a negative result about plasticity: a coupling update descending
39 a squared-drift objective preserves coherence while moving *away* from the

environmental structure it is expected to learn, below a shuffled-label comparison. Dissipation minimization does not imply representational learning. The fourth is a machine-checked composition (Section 5) in which finite information capacity, Landauer’s bound, a predictive dissipation bound, the noisy Kuramoto transition, sheaf gluing and Banach’s fixed-point theorem yield one conditional theorem, with the eight hypotheses that composition consumes isolated and named; a common toy model satisfies all eight, so their conjunction is not empty.

A prose table of assumptions would list the same eight items, so the fourth contribution needs a word about method. What the Lean development adds is that it makes the list non-negotiable. The characteristic failure in an argument of this shape is not a false step but a silent change of subject: a theorem about a stationary density read as a claim about a trajectory, an inequality about a register read as a claim about learning, agreement of phases read as compatibility of representations. Type-checking forbids each of those substitutions, and an assumption needed to make a step go through appears as a hypothesis in the term rather than as a word in a sentence. The last of them is not merely forbidden but refuted by a constructed object: the development carries a cover at a single common phase whose local densities disagree where its patches meet, satisfying every other requirement of the gluing theorem and failing compatibility alone (Section 5.2). That object is what a prose table of assumptions cannot carry. Formalization does not make the assumptions true; it makes them impossible to lose.

63 The physical model and the formal composition belong in one article be-
64 cause they are two halves of one commitment. The coupling-gated prediction
65 is worth testing under this framework, rather than as one more synchroniza-
66 tion model, only because the coherent state it produces is claimed to do
67 further work: to support a globally consistent representation and a reflexive
68 fixed point that we identify with phenomenal unity and a minimal self.

69 The distinction that does that work is between phases agreeing and de-
70 scriptions fitting together, and it is concrete. Take two cortical regions whose
71 receptive territories overlap, each carrying a probability distribution over
72 what occupies the shared part. Their oscillators can be locked to the same
73 phase at every instant while one region's distribution on the overlap places an
74 object left of fixation and the other places it right. The phases agree; the de-
75 scriptions do not, and no single distribution over the shared territory restricts
76 to both. A theory that identifies unity with synchrony cannot mark that dif-
77 ference, and so cannot separate a brain whose regions oscillate together from
78 one whose regions are about the same world. The sheaf condition marks
79 exactly it: local states glue to one global section when their restrictions to
80 shared regions coincide, and otherwise not at all. Coherence enters as the
81 physical condition under which compatibility can be maintained, not as a
82 substitute for it, and the self-prediction map then asks the further ques-
83 tion of whether the glued state contains an accurate model of itself. That
84 identification is a commitment, not a theorem, and Section 6 states it, ar-
85 gues for it, and says what would count against it. Its separation from the

physical tests is what makes both assessable: the cortical coupling hypothesis and the recovery prediction can fail while the mathematical composition stands, and the composition can stand while the identification is wrong. The established ingredients (Landauer, 1961; Still et al., 2012; Sakaguchi, 1988; Banach, 1922) therefore appear here for their biological composition, not for any new fundamental physical law.

2 A coupling-gated model of cortical coherence

2.1 Dynamics and stationary solutions

The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

Here $D > 0$ is phase diffusion, W_i are independent Wiener processes, and K_{ij} has inverse-time units, with angles treated as dimensionless. The uniform mean-field convention is $K_{ij} = K/N$. Frequency heterogeneity and phase diffusion are separate parameters; a frequency-spread width cannot be substituted for D when applying the identical-frequency stationary theory.

For identical frequencies in their rotating frame, the assumed stationary

102 density is

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}. \quad (2)$$

103 The I_n are modified Bessel functions, and ψ is the collective phase. Lean
 104 verifies normalization, the density's circular order parameter, and the self-
 105 consistency equation. At $K \leq K_c = 2D$, the only nonnegative self-consistent
 106 order parameter is zero. At $K > K_c$, exactly one positive solution coex-
 107 ists with zero; the positive branch is continuous at threshold and strictly
 108 increasing with K .

109 The near-threshold expansion gives

$$\frac{r^2}{K - K_c} \rightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

110 This is a result about the stationary solution branch. Of the coherent branch
 111 we prove existence, uniqueness, continuous emergence at threshold and strict
 112 growth in K , but not that it is dynamically selected. The dynamical mean-
 113 field limit—propagation of chaos—is the missing link between the finite sys-
 114 tem and the stationary description (Sznitman, 1991), and no theorem in this
 115 development closes it. Exchange symmetry alone does not justify factoriza-
 116 tion of the joint law, particularly for an ensemble with an unpinned collective
 117 phase.

118 A spatial version replaces the finite sum by an integral against a kernel
 119 $K(x, y)$. We normalize the substrate measure to total mass one, so the mean-

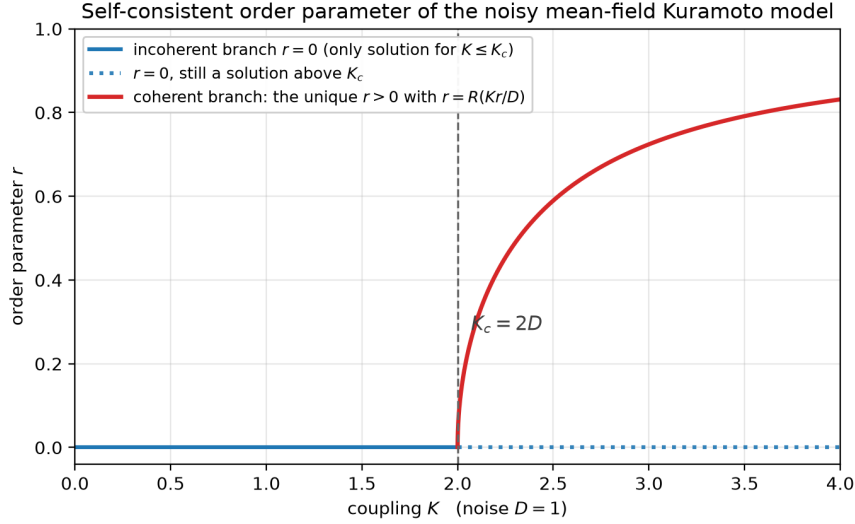


Figure 1: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the same integral expressions used in the formalization; it illustrates the static theorems and does not establish dynamical selection.

field strength is a kernel average, not a quantity that grows with the arbitrary
measure of the substrate. Reduction of a heterogeneous spatial kernel to a
scalar K is an additional modelling step, and the threshold in Eq. (2) does
not automatically apply to every spatial or signed network.

2.2 Physical identification and calibration

Regular mesh refinement gives convergence of a discrete coupling-energy
quadrature to its continuum integral: scalar energy convergence, not conver-
gence of trajectories and not a construction of a physical kernel. The model

128 posits a continuum kernel and separately identifies its cortical realization
 129 with endogenous electromagnetic coupling. The formal predicate requires
 130 a jointly continuous kernel, a normalized substrate, a mean-field strength
 131 equal to its spatial average, and no single site providing more than half of
 132 that mean-field coupling. Toy witnesses satisfy these conditions; none is a
 133 model calibrated to cortex.

134 Extracellular fields can modulate spike timing (Fröhlich & McCormick,
 135 2010; Anastassiou & Koch, 2015), and field-based models have been proposed
 136 for integrating distributed neural information (Pinotsis & Miller, 2022; Mc-
 137 Fadden, 2020). These observations motivate an effective interaction without
 138 establishing its magnitude relative to phase diffusion, its spatial averaging
 139 rule, or its causal role in experience; changes in phase organization under
 140 propofol (Bastos et al., 2021) likewise do not measure a crossing of the model’s
 141 coupling threshold.

142 The physical-unit conversion is the sharpest open limitation of the frame-
 143 work. An estimate proportional to $N\Delta tf$, using population size, a spike-time
 144 shift and oscillation frequency, is dimensionless. It cannot be compared di-
 145 rectly with an inverse-time diffusion parameter. A conversion such as

$$K_{\text{eff}} = \gamma N s E f \tag{4}$$

146 requires a rate calibration γ , where s is the time shift per unit field and E the
 147 field amplitude. The saved field-strength calculations are conditional on this

148 calibration and on the chosen spatial aggregation; they do not independently
 149 establish $K_{\text{eff}} > 2D$ in cortex. Determining the interaction kernel and phase
 150 diffusion from the same physical system is necessary before the threshold
 151 becomes a quantitative biological claim; Section 3.2 makes that measurement
 152 the first item of the proposed experiment.

153 **3 Recovery after awakening: a conditional** 154 **prediction**

155 Sleep-associated changes in extracellular space and clearance motivate a pos-
 156 sible slow control of coupling (Xie et al., 2013). We hypothesize that recovery
 157 after awakening includes a change in effective coupling $K(t)$, mediated by
 158 extracellular geometry, on a slower timescale than phase dynamics. Neither
 159 the monotonic relation between geometry and coupling nor its role in hu-
 160 man sleep inertia follows from the cited physiological observation; the model
 161 requires both to be measured.

162 If $K(t)$ rises through K_c and the phase distribution tracks the coherent
 163 stationary branch, Eq. (3) predicts

$$r(t) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad t > t_0, \quad \dot{K}(t_0) > 0. \quad (5)$$

164 The ideal stationary description has a subthreshold interval followed by con-
 165 tinuous onset. Saturation requires a specified subsequent coupling trajectory;

monotonic coupling alone does not determine a logistic or any unique sigmoid. The onset time is fixed by $K(t_0) = K_c$ only if $K(t)$ is independently known. When coupling is inferred from the same coherence trace, this is not a parameter-free prediction of latency.

Critical slowing down limits the adiabatic prediction of Eq. (5) near threshold, and a finite population carries nonzero fluctuating order below it. An observable onset therefore requires a predeclared threshold above the finite-size floor, and a finite-rate trajectory need not show an infinite-slope cusp. Section 4.1 quantifies the limitation.

3.1 Distributional and temporal tests

The stationary model makes a second prediction: phase concentration and coherence obey $r = I_1(a)/I_0(a)$, with no fitted shape parameter. Its applicability depends on the stationary-density assumption and the observation model, and the relation is common to the whole von Mises family, so agreement with it identifies neither ephaptic coupling nor this framework against alternative synchronization models. Within the same ansatz $a/r = K/D$ for $r > 0$, so the limiting threshold ratio follows from the density relation instead of testing it, and ratios near the finite-size floor are especially unstable.

The estimators must be independent, which is the constraint most easily violated in practice. Estimating a by inverting the Bessel ratio from r would enforce the relation being tested, and linear statistics on wrapped phases estimate neither quantity.

188 **3.2 A protocol for the awakening experiment**

189 The requirements above define a design rather than a single measurement,
190 and the design is where the content of the hypothesis lives. An adequate test
191 needs all six of the following.

- 192 **1. Coupling and diffusion in the same units.** Estimate K and D
193 from the same cortical regime, in inverse-time units, separating phase
194 diffusion from quenched frequency spread and from observation noise.
195 Without this the threshold $K > 2D$ is not an evaluable statement
196 about cortex.
- 197 **2. Geometry calibrated to coupling.** A diffusion-MRI observable or
198 an astrocytic signal is an indicator of extracellular geometry, not a mea-
199 surement of K . Its mapping to effective coupling must be established
200 separately, with its own uncertainty.
- 201 **3. Phase estimation at adequate spatial resolution.** At least 100
202 spatial samples per estimate. Estimate r by the circular resultant
203 length and a by an unweighted regression of log-density on the co-
204 sine of phase relative to the circular mean. With fewer sites, pooling
205 over time changes the estimand to a spatiotemporal distribution and
206 must be stated as such.
- 207 **4. An independent recovery endpoint.** Behavioral recovery measured
208 separately from the phase trace, so that latency is not defined by the

209 signal whose latency is at issue.

210 **5. A predeclared onset criterion.** A coherence threshold fixed above
211 the finite-size floor before unblinding, chosen in the knowledge that a
212 finite ramp smears the cusp.

213 **6. Model comparison under matched observation.** Coupling-gated
214 recovery compared against exponential, flexible sigmoidal and other
215 slow-recovery models under identical temporal filtering and observation
216 noise. Latency alone cannot distinguish the mechanism, because many
217 processes create delays.

218 The design fails informatively in three ways: failure of the stationary
219 phase-density relation over a sufficiently broad concentration range; absence
220 of the proposed coupling–geometry association; or a recovery trace better
221 explained by an alternative after finite-rate and observation effects are ac-
222 counted for. Two apparent failures are not failures. A linear apparent onset
223 is evidence against the adiabatic approximation only where that approxima-
224 tion is independently justified, and it does not falsify the field hypothesis
225 under arbitrary ramp speeds. The absence of a fold in the stationary equa-
226 tion excludes stationary coexistence of two positive branches in this model,
227 but does not exclude finite-rate hysteresis or adjudicate theories of ignition.

228 4 Simulation results

229 The simulations characterize the finite-size, finite-rate regime that the sta-
230 tionary theory leaves open, and they bear on whether the prediction of Sec-
231 tion 3 is measurable at all. Two of them, Sections 4.1 and 4.3, are re-
232 sults about method: they hold for any analysis of a driven transition or of
233 dissipation-driven plasticity, whatever one makes of the field account. Each
234 is a finite run at its sampled parameters and not a general statement, and
235 none discharges a hypothesis of the composition in Section 5. Each run uses
236 explicit seeds and compact saved summaries, and the numerical values be-
237 low are generated from those artifacts. The supplement gives parameters,
238 estimators, failure criteria and additional panels. Identical-frequency runs
239 use phase diffusion rather than a quenched frequency-spread surrogate, and
240 order plots display the $1/\sqrt{N}$ reference.

241 4.1 Ramp speed sets the apparent onset exponent

242 The dynamic-ramp study (Fig. 2) uses $N = 2000$, $D = 1$ and 32 replicas at
243 four coupling speeds. Escape is the first of three consecutive saved samples at
244 $r \geq 0.2$ after threshold. Only 5/32 replicas escape within the fastest window.
245 The three slower ramps have mean coupling delays 0.1947, 0.0684 and 0.0253,
246 with a fitted delay exponent 0.443. The apparent onset exponent changes
247 from 0.964 at the fastest speed to 0.417 at the slowest.

248 The exponent recovered by fitting a driven transition is a joint function

249 of the mechanism and the drive rate. An onset exponent read off a recovery
 250 trace therefore says little about the underlying transition unless the ramp rate
 251 is independently known and the finite-size floor is declared, and that holds
 252 for any such analysis, not only for one conducted under this framework. The
 253 prediction of Section 3 is the first thing it constrains: the stationary square-
 254 root law cannot be imposed on an arbitrary recovery trace, and its absence
 255 is not by itself evidence against a coupling-gated mechanism. The specified
 256 deviation-based critical ramp speed is not bracketed by the sweep.

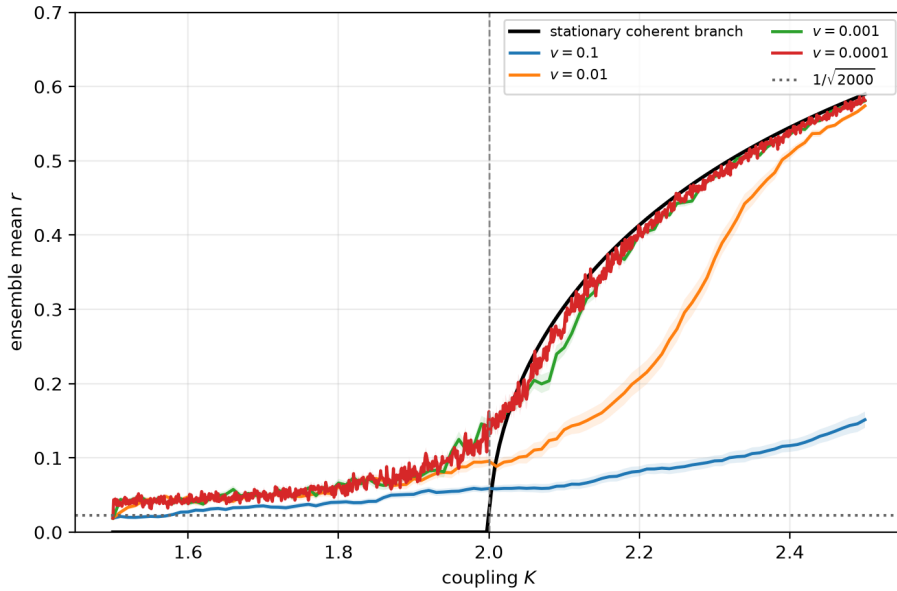


Figure 2: Finite-rate departures from the stationary branch. Ensemble-mean order is plotted against the increasing coupling for four ramp speeds v , with bands of one standard error across replicas. The dashed vertical line is the static threshold and the dotted horizontal line the finite-size reference. Faster ramps delay escape; the stationary branch is not a universal time course for recovery.

257 The dynamical-selection control shows escape toward a coherent branch
 258 above threshold, but retains quantitative finite-size discrepancies. At $K =$
 259 2.8, steady mean order is 0.67735 against the static value 0.68270. The
 260 fitted growth-rate slope is 0.43240, compared with the theoretical slope $1/2$.
 261 The propagation-of-chaos diagnostics show decreasing subcritical dependence
 262 with increasing population size. In the supercritical unpinned ensemble,
 263 however, pair correlation remains substantial because realizations share a
 264 random collective orientation. This negative unconditional result does not
 265 settle a conditional mean-field limit after removal of that orientation.

266 **4.2 Spatial coupling, frustration and calibration**

267 A periodic two-dimensional sheet with row-normalized exponential coupling
 268 remains coherent in the sampled empirical decay-length band under the de-
 269 clared total coupling and frequency spread. At decay lengths 0.1, 0.2 and
 270 0.3 mm, steady order is 0.9424, 0.9427 and 0.9427. This is one finite param-
 271 eter regime; fixing total coupling while changing range tests fragmentation,
 272 not whether cortex supplies that total coupling.

273 A row-balanced Dale-law network shows that row balance alone does not
 274 guarantee a floor-level incoherent baseline. In a separately specified weak-
 275 synapse regime, a uniform modulatory term produces an operational coher-
 276 ence crossing at K_{eff}/D between 1.8994 and 1.9387. That ratio is dimension-
 277 less; carrying it over to cortex needs the rate calibration γ of Section 2.2,
 278 which neither this sweep nor the spatial control supplies.

279 4.3 Squared-drift plasticity does not learn structure

280 A coupled system that descends a dissipation-like objective is widely expected
 281 to come, in doing so, to mirror the structure of its input; the predictive
 282 dissipation bound of Section 5 invites that reading. This control tests the
 283 expectation directly in the present model, and the expectation fails.

284 For fixed phases, consider the squared-drift objective

$$\sigma(K) = \frac{1}{D} \int \left[\omega(x) + \int K(x, y) \sin(\theta_y - \theta_x) d\mu(y) \right]^2 d\mu(x). \quad (6)$$

285 It is a quadratic functional of an affine image of the kernel. The formalization
 286 constructs the bounded drift operator A and its adjoint on the relevant L^2
 287 spaces, obtaining $\nabla\sigma = (2/D)A^*(AK + \omega)$, and the specified gradient flow
 288 decreases this objective at fixed phases. The objective is not generally a
 289 thermodynamic entropy-production rate, which requires probability-current
 290 information. Whether biological couplings follow this flow is an additional
 291 modelling question.

292 The joint control evolves noisy phases with clustered frequencies and
 293 nonzero grand-mean drive, while updating symmetric couplings every 50 fast
 294 steps. Couplings remain nonnegative, hollow and normalized to a fixed to-
 295 tal. The symmetric-edge derivative of $\frac{1}{2} \sum_i v_i^2$ is $(v_i - v_j) \sin(\theta_j - \theta_i)$, so both
 296 endpoints contribute. Resource constraints and evolving phases remove the
 297 fixed-phase monotonicity guarantee.

298 Across three seeds, second-half order lies between 0.9811 and 0.9815, and

299 matched frozen-coupling controls maintain similar coherence. The distance
300 to the true cluster template increases in every adaptive run, with worse
301 changes than the shuffled-label comparison (Fig. 3). The template is a co-
302 variance of shared latent cluster signals, not an input to the update or a
303 measured covariance of the constant frequencies.

304 The objective descends, coherence is maintained, and the coupling matrix
305 moves away from the generating structure—below the level of a random
306 relabelling. Coherence is therefore not a proxy for representational alignment
307 in this regime, and a plateau in the objective establishes neither stationarity
308 of the adaptive process nor a thermodynamic entropy-production rate.

309 4.4 The estimation protocol demonstrated on public 310 EEG

311 Item 3 of Section 3.2 specifies estimators whose behavior on real recordings
312 is not obvious, so we exercise them on public propofol-sedation EEG (Bajwa
313 et al., 2025). The purpose is to show what the protocol produces and where
314 it is fragile, not to test the awakening hypothesis, for which these data are
315 the wrong design.

316 Two fragilities appear immediately. The implemented estimator pools 62
317 channels over a 100-ms window, so it estimates a spatiotemporal distribu-
318 tion rather than an instantaneous spatial one—a departure from the protocol
319 forced by channel count, and one that any scalp implementation will face.

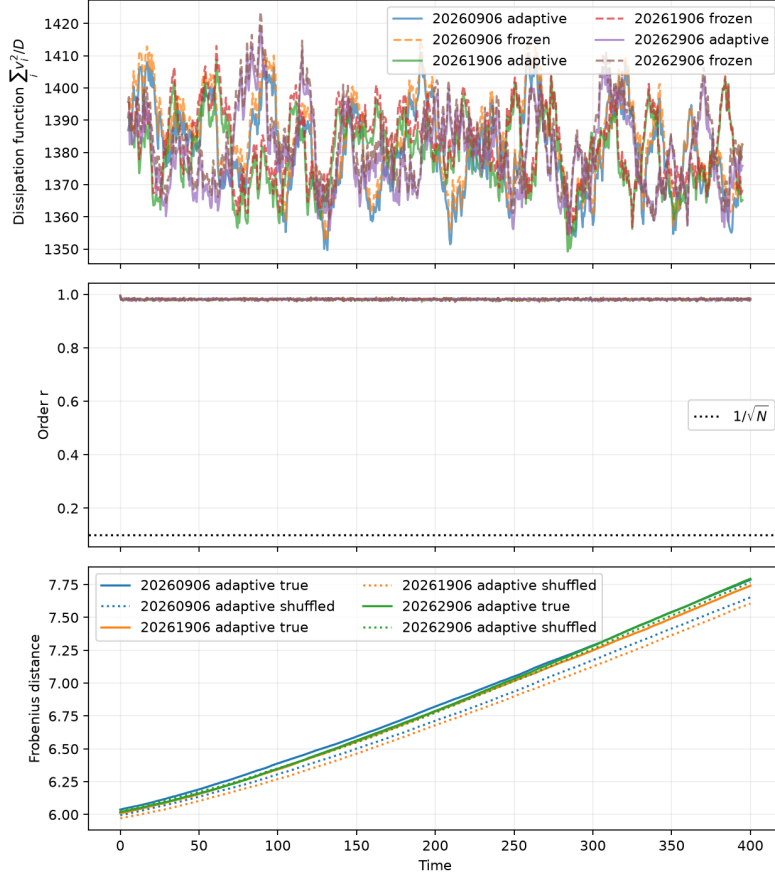


Figure 3: Joint fast/slow dynamics with matched frozen-coupling controls. Top: smoothed squared-drift dissipation, with frozen controls dashed. Middle: order and the finite-size reference. Bottom: distances to true cluster templates (solid) and node-label shuffles (dotted). Coherence persists, but structural distances increase. The result limits the proposed connection between dissipation minimization and environmental structure.

320 Bipolar and common-reference montages give different residuals, making ref-
 321 erencing a material analysis choice rather than a preprocessing detail.

322 The bipolar results are compatible with the Bessel relation over the ob-
 323 served concentration range. That range is too narrow to distinguish the
 324 curve from its linear approximation at the observed scatter, so the check is
 325 uninformative about the relation itself. An informative test needs a broader
 326 concentration range and explicit treatment of source mixing, finite samples
 327 and temporal pooling; intracranial or high-density recordings across a gen-
 328 uine state transition are the natural setting.

329 5 Thermodynamic constraints and the for- 330 mal composition

331 The synchronization model of Section 2 is one link of a longer conditional
 332 construction, whose nine nodes are:

333 **n1** a finite phase space has bounded information capacity; **n2** a
 334 broken symmetry forces a boundary, where the field leaves its vac-
 335 uum manifold; **n3** the boundary erases, and erasure costs heat;
 336 **n4** that dissipation bounds nonpredictive memory; **n5** discrete
 337 couplings coarse-grain to a continuum limit; **n6** a specified phys-
 338 ical field realizes that limiting coupling; **n7** above threshold the
 339 field carries a unique coherent order parameter; **n8** compatible
 340 local states glue to a unique global section; **n9** that section is the
 341 unique fixed point of a self-prediction map.

342 Node-level results are established or cited; what the composition consumes

are the eight implications between them. Figure 4 shows where they sit and what kind of assumption each one is, and Table 1 states them, classified by their role in the composition rather than by their plausibility. In particular, finite capacity does not imply a symmetry-breaking boundary: n_1 and n_2 are independent roots, even though the composition can accept an implication between them as an argument.

5.1 Capacity, boundaries and predictive memory

For a finite state space X , Shannon entropy satisfies $H(p) \leq \log |X|$, with equality only for the uniform distribution. A stream with more possible histories than $|X|$ cannot be encoded injectively in that register. Separately, a continuous field on a connected substrate that reaches distinct components of a disconnected vacuum manifold must leave that manifold somewhere, the simplest topological-defect construction (Kibble, 1976). Relating this locus to a particular finite information-processing register is the additional hypothesis E12 carries.

A non-surjective update on a finite register is non-injective. Under the specified system–bath model and thermal assumptions, erasure carries a Landauer heat cost (Landauer, 1961). Finite capacity does not imply that every update erases, and logical many-to-one processing must be distinguished from its physical implementation (Norton, 2005; Shenker, 2000). The formalization records the physical heat identity as an obligation of each model and supplies an explicit one-bit eraser satisfying it.

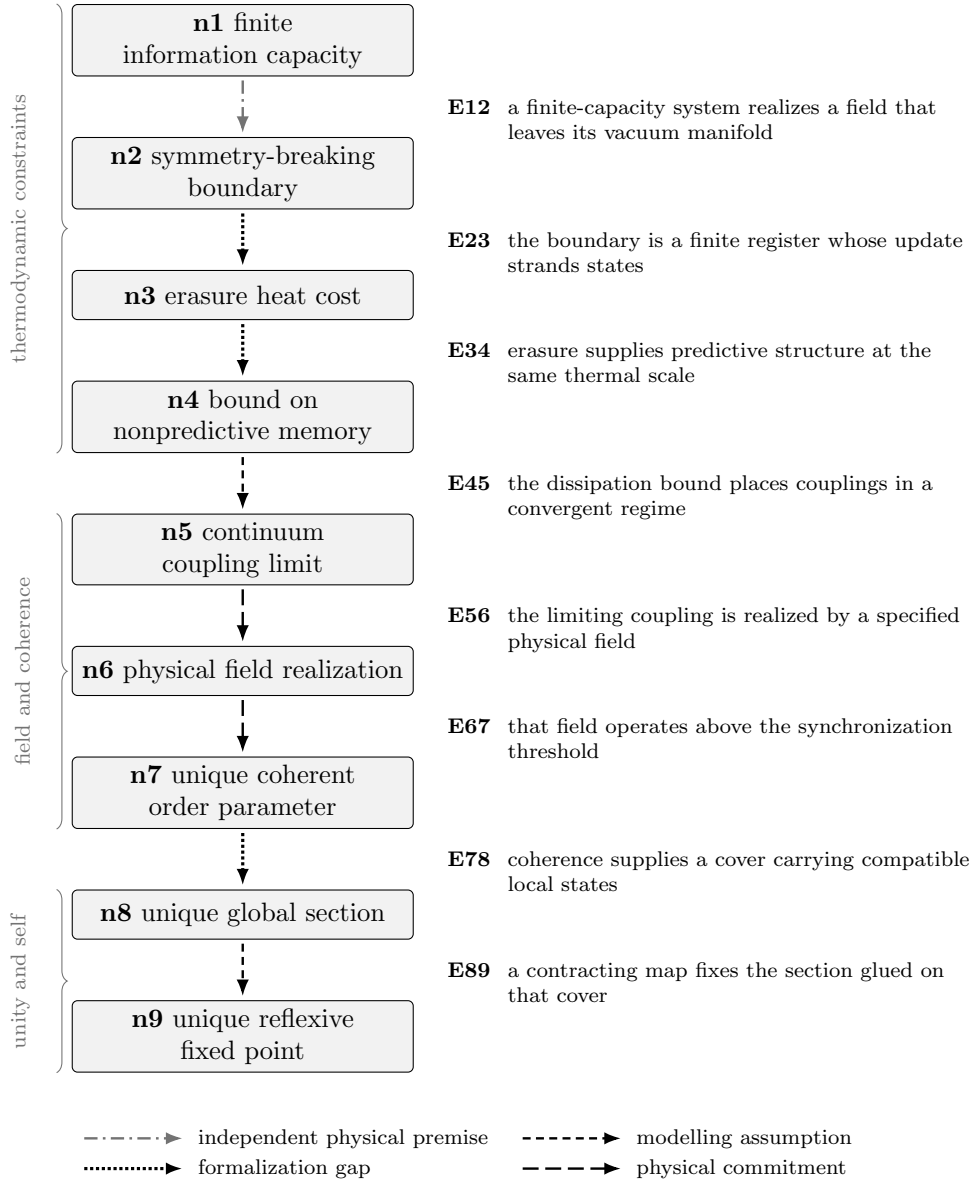


Figure 4: The conditional composition. Each node is an established or cited result; each arrow is an implication the composition consumes and no theorem here supplies, styled by the kind of assumption it carries. The three formalization gaps bracket the two physical commitments, so the chain’s empirical content sits between links of a different kind on either side. E12 is drawn as a premise because n1 and n2 are independent roots. Table 1 states each connection in full.

Edge	Status	Required connection
E12 capacity $\rightarrow$ boundary	Independent physical premise	A finite-capacity system realizes a field that leaves its vacuum manifold. Capacity alone does not imply this.
E23 boundary $\rightarrow$ erasure	Formalization gap	The boundary is associated with a finite register whose update leaves states unreachable.
E34 erasure $\rightarrow$ prediction	Formalization gap	Erasures supplies a predictive structure with the same thermal scale and heat budget, and nonzero wasted memory.
E45 prediction $\rightarrow$ continuum	Modelling assumption	A predictive dissipation bound places coupling energies in a convergent coarse-graining regime.
E56 continuum $\rightarrow$ field	Physical commitment	The limiting coupling is realized by a specified field satisfying the kernel conditions. Its identification with cortex is empirical.
E67 field $\rightarrow$ coherence	Physical commitment	That field operates above the synchronization threshold.
E78 coherence $\rightarrow$ unity	Formalization gap	Coherence supplies a cover with coupling at least the named mean-field value, reached by relaxation and carrying compatible local states.
E89 unity $\rightarrow$ self	Modelling assumption	The state glued on that same cover is fixed by a contracting map mediated by its local self-encoding.

Table 1: The assumptions consumed by the conditional composition theorem, each an implication between two of the nine nodes above. Node-level results do not establish these connecting implications. The complete theorem map and witnesses are in Supplemental Material, Table S1.

365 For a system state X_t driven by an environmental signal S_t , the predictive
 366 dissipation bound is

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (7)$$

367 Under the no-back-action Markov structure $X_t \rightarrow S_t \rightarrow S_{t+1}$, data process-
 368 ing makes the bracket nonnegative (Still et al., 2012). The bound constrains
 369 nonpredictive memory at a given dissipation budget. It does not specify
 370 a learning rule, maximize prediction, or require the internal representation
 371 to reproduce environmental geometry—the point the plasticity control of
 372 Section 4 makes concretely. Connecting this budget to Landauer heat ad-
 373 ditionally requires nonpredictive information to be bounded by the entropy
 374 erased by the register. The formal bridge derives Eq. (7) from Landauer’s
 375 bound under that identification; it does not derive the identification.

376 A bound $D_{KL} \leq \Delta t \sigma$ also cannot force $D_{KL} \rightarrow 0$ unless its right-hand
 377 side tends to zero, and Gibbs’ inequality proves nonnegativity of relative en-
 378 tropy, not a general identification of relative entropy with physical entropy
 379 production. These distinctions prevent an informational inequality from be-
 380 ing read as a derivation of environmental learning, Bayesian inference, or
 381 semantic content.

382 5.2 Compatible states and reflexive fixed points

383 A sheaf formalizes local descriptions and their restriction to smaller regions.
384 If local sections agree on every overlap, the sheaf condition gives a unique
385 global section. In the probability-sheaf construction, this is consistency of
386 local density information, and it is not implied by agreement of oscillator
387 phases: the development includes a counterexample with equal phases but
388 incompatible densities on an overlap. Section 6 rests on this distinction,
389 and the model excludes such configurations only through its compatibility
390 hypothesis.

391 A thermodynamic cover couples this gluing construction to a phase con-
392 figuration at a potential minimum. For the deterministic positive-coupling
393 model, convergence to that minimum is proved under an explicit initial basin
394 condition; other equilibria prevent an unconditional conclusion. The station-
395 ary noisy order parameter, the deterministic relaxation, and the cover must
396 therefore be connected by E78 rather than identified without a hypothesis.
397 Neither field continuity nor synchronization alone guarantees a globally com-
398 patible representation.

399 A reflexive extension equips global sections with a complete metric and a
400 predictive map formed by restricting a state to a local encoding region and
401 reading that encoding out. If the map contracts and fixes the section supplied
402 by the cover, Banach's theorem gives uniqueness of that same fixed point.
403 The contraction rate is assumed, and the equality between the glued section
404 and the fixed point is part of E89. A coherent order parameter supplies the

405 supercritical inequality required by the stated contraction bound; it does not
406 construct the map or prove that cortical dynamics implements it.

407 The theorem `chain` concludes `UnifiedSelf` from these hypotheses. A
408 common toy witness satisfies all eight, so their conjunction is not empty.
409 The project declares no additional Lean axioms beyond the foundational
410 axioms used by Lean’s libraries; physical content resides in hypotheses and
411 model fields. This checks the conditional mathematical composition, not the
412 empirical truth of its premises. Detailed constructions, scope statements and
413 counterexamples are retained in the supplement and source files.

414 6 What the construction commits to about 415 consciousness

416 The composition terminates in a mathematical object, and we take a posi-
417 tion on what that object is. The commitment has two clauses. *Unity*: the
418 unique global section glued from compatible local states is the formal corre-
419 late of the unity of a conscious episode—the fact that a subject undergoes
420 one experience rather than a collection of co-occurring local ones. *Selfhood*:
421 the fixed point of the self-prediction map on that same section is the formal
422 correlate of a minimal self, a state that is what it takes itself to be. Neither
423 clause is a theorem, and no simulation here supports either. We argue for
424 them, then say what would count against them.

425 Three features of the construction motivate the identification, and each

distinguishes it from a bare synchrony measure.

First, compatibility rather than agreement. Oscillator phases can agree everywhere while the local probabilistic descriptions they carry remain incompatible on their overlaps, and Section 5.2 exhibits that configuration inside the formalization, not only as an illustration. If the unity of experience is a matter of local contents being about one world rather than merely co-timed, then compatibility, not coherence, is the property to look for—and coherence enters as the physical condition under which compatibility can be maintained, which is what E78 asserts.

Second, uniqueness follows from the construction rather than from a separate selection principle. Accounts of conscious unity need something that picks one experience from among the candidates a system could support; integrated information theory postulates exclusion for this purpose (Tononi et al., 2016). Here uniqueness arrives twice as a consequence: given a cover whose local sections agree, the sheaf condition gives exactly one global section, and given a contracting self-prediction map, Banach’s theorem gives exactly one fixed point. The scope of that claim needs stating exactly, because the stronger reading is available and is not ours. Uniqueness is relative to a cover and a map, and choosing them—which regions, which overlaps, which encoding region, and so where the subject ends—is not supplied by either theorem. It is what E78 and E89 carry. What the construction removes is the need for an additional principle to select among the experiences a fixed cover and map would otherwise admit; it does not remove the need to say

449 which cover and map a given subject has. That is a structural argument,
450 not evidence.

451 Third, the reflexive clause makes the self a condition rather than an addi-
452 tion. The predictive map restricts a global state to a region encoding it and
453 reads that encoding back out, so a fixed point is a state whose self-model is
454 accurate about the state it models. That condition is available only because
455 the state modelled is a single glued object: a system whose local descrip-
456 tions do not cohere has no such state for a self-model to be about. Unity
457 and selfhood are therefore one construction read at two levels, which is why
458 `UnifiedSelf` requires the fixed point to be the glued section itself rather
459 than an unrelated section.

460 The costs belong with the argument. The commitment does not address
461 why any physical condition is accompanied by experience; it is a proposal
462 about which physical state is the conscious one, not about why states of that
463 kind feel like anything. It is therefore open to the objection that a system
464 could satisfy every condition—compatible local states, a supercritical field,
465 an accurate self-model—while nothing is experienced. We have no reply to
466 that objection and claim none. The identification is also multiply realizable
467 as stated: nothing in the sheaf or fixed-point conditions is specific to cortex
468 or to electromagnetic coupling, so the substrate claim is carried entirely by
469 E56 and E67 and is empirical. No conclusion about artificial systems follows
470 from the formal results.

471 What could count against the commitment is dissociation. It predicts that

472 the two clauses and the presence of a unified conscious episode go together,
473 so a state in which local cortical descriptions are demonstrably compatible
474 and the field supercritical, while the subject reports no unified experience,
475 would tell against it; so would unified reported experience where compati-
476 bility demonstrably fails. Anesthetic emergence, dreaming and dissociative
477 states are the places to look, because they separate coherence from report
478 in ways ordinary waking does not. Such tests need an operational measure
479 of overlap compatibility on neural data, which this article does not supply;
480 constructing one is the main obstacle to making the commitment empirically
481 live rather than merely stated.

482 The physical tests of Sections 3 and 4 are independent of all of this. They
483 concern coupling, coherence and recovery, and can fail while the identification
484 stands, or succeed while it is wrong.

485 **7 Discussion**

486 The framework connects thermodynamic information constraints, collective
487 phase organization and compatible local representations, conditionally and
488 with its assumptions named. Its biological content is a coupling-gated re-
489 covery hypothesis with defined observables, a stated protocol, and failure
490 conditions declared in advance. Its formal content is a composition whose
491 eight gaps are exhibited rather than argued past, and which can be inhabited
492 by a single model. Its interpretive content is the identification of Section 6,

493 offered as a commitment with reasons.

494 The simulations are useful because they restrict the hypothesis rather
495 than support it. Finite ramps smear the stationary onset to the point where
496 a fitted onset exponent chiefly reports the ramp; an unpinned collective
497 phase makes independence diagnostics ambiguous in exactly the supercritical
498 regime the model cares about; and squared-drift plasticity maintains coher-
499 ence while degrading structural alignment. The second and third of these
500 are results about method and about a common intuition, and they transfer
501 beyond this framework.

502 The proposal differs in emphasis from accounts based on integrated in-
503 formation (Tononi et al., 2016), global workspace dynamics (Baars, 1997;
504 Dehaene et al., 2014), or predictive processing (Friston, 2010). It specifies a
505 candidate interaction medium and a mathematical compatibility condition,
506 and, relative to a given cover and self-prediction map, it derives the unique-
507 ness those accounts postulate. It does not derive the exclusion principle of
508 integrated information, identify every broadcast with a field transition, or
509 derive a generative model from a dissipation bound. Continuous fields can
510 carry incompatible local descriptions and discrete models can admit sheaf
511 constructions, so continuity alone does not settle the combination problem;
512 compatibility is what this account offers in its place.

513 Two formal results bear on the substrate question, and both admit a
514 stronger reading than they support. At matched coupling resource, a fixed
515 wiring support that excludes a better-correlated pair can be suboptimal—a

516 statement about constrained optimization, not about biology. On an atom-
517 less substrate, a kernel supported on finitely many points contributes zero to
518 the chosen continuum coupling integral. That is a property of the measure
519 and the observable, and it cuts against this framework’s own coarse-graining
520 step as much as against anything else. It does not imply that digital hard-
521 ware cannot approximate a field or support consciousness, and no biological-
522 versus-artificial conclusion follows.

523 The empirical priorities follow directly from Section 3.2: estimate effec-
524 tive coupling and diffusion in consistent units, test the proposed relation
525 between coupling and extracellular geometry, and measure recovery against
526 an independently defined behavioral endpoint under matched model compar-
527 ison. A favorable concentration–coherence relation would support a statis-
528 tical description, not identify its physical cause; an unfavorable result in a
529 regime satisfying the stated assumptions would constrain that description.
530 Successful synchronization simulations cannot substitute for measurements
531 distinguishing ephaptic interactions from other coupling mechanisms.

532 The article therefore asks to be judged in three separable parts: a pre-
533 diction that can be tested, a composition that can be checked, and an iden-
534 tification that can be argued about. Keeping them separate is what makes
535 the whole assessable.

536 **Data and code availability**

537 The Lean development, simulation scripts, tests and saved numerical sum-
538 maries are maintained in the manuscript source repository. Publication sim-
539 ulation values are regenerated from saved summaries without rerunning pro-
540 duction sweeps. The exploratory EEG analysis uses OpenNeuro dataset
541 ds005620 (Bajwa et al., 2025); preprocessing and selection are described
542 in Supplemental Material. The source repository contains the reproducible
543 analysis and does not redistribute the raw recordings.

544 **Use of AI tools**

545 AI tools assisted simulation-code development and manuscript revision. Ma-
546 chine checking establishes the stated Lean results under their hypotheses;
547 it does not validate biological interpretations or replace scientific review of
548 generated code and prose.

549 **References**

- 550 Pinotsis, D. A., & Miller, E. K. (2022). Beyond dimension reduction: Stable
551 electric fields emerge from and allow representational drift. *NeuroImage*,
552 253, 119058.
- 553 Tononi, G., Boly, M., Massimini, M., & Koch, C. (2016). Integrated informa-

tion theory: from consciousness to its physical substrate. *Nature Reviews*
Neuroscience, 17(7), 450-461.

Baars, B. J. (1997). In the Theater of Consciousness: The Workspace of the
Mind. *Oxford University Press*.

Dehaene, S., Charles, L., King, J. R., & Marti, S. (2014). Toward a compu-
tational theory of conscious processing. *Current Opinion in Neurobiology*,
25, 76-84.

Friston, K. (2010). The free-energy principle: a unified brain theory?. *Nature*
Reviews Neuroscience, 11(2), 127-138.

Pinotsis, D. A., & Miller, E. K. (2023). In vivo ephaptic coupling allows
memory network formation. *Cerebral Cortex*, 33(17), 9877-9890.

Pinotsis, D. A., Fridman, G., & Miller, E. K. (2023). Cytoelectric coupling:
Electric fields sculpt neural activity and "tune" the brain's infrastructure.
Progress in Neurobiology, 226, 102465.

McFadden, J. (2002). The conscious electromagnetic information (cemi) field
theory: the hard problem made easy?. *Journal of Consciousness Studies*,
9(8), 45-60.

Pockett, S. (2012). The electromagnetic field theory of consciousness: a
testable hypothesis about the characteristics of conscious as opposed to
non-conscious fields. *Journal of Consciousness Studies*, 19(11-12), 191-223.

574 McFadden, J. (2020). Integrating information in the brain's EM field: the
575 cemi field theory of consciousness. *Neuroscience of Consciousness*, 2020(1),
576 niaa016.

577 Fröhlich, F., & McCormick, D. A. (2010). Endogenous electric fields may
578 guide neocortical network activity. *Neuron*, 67(1), 129-143.

579 Anastassiou, C. A., & Koch, C. (2015). Ephaptic coupling to endogenous
580 electric field activity: why bother?. *Current Opinion in Neurobiology*, 31,
581 95-103.

582 Pockett, S. (2002). Difficulties with the electromagnetic field theory of con-
583 sciousness. *Journal of Consciousness Studies*, 9(4), 51-56.

584 Vöröslakos, M., et al. (2018). Direct effects of transcranial electric stimulation
585 on brain circuits in rats and humans. *Nature Communications*, 9, 483.

586 Verkhratsky, A., & Nedergaard, M. (2018). Physiology of Astroglia. *Physio-*
587 *logical Reviews*, 98(1), 239-389.

588 Bastos, A. M., Donoghue, J. A., Brincat, S. L., Mahnke, M., Yanar, J., Cor-
589 rea, J., ... & Miller, E. K. (2021). Neural effects of propofol-induced uncon-
590 sciousness and its reversal using thalamic stimulation. *eLife*, 10, e60824.

591 Bajwa, I. J., Nilsen, A. S., Skukies, R., Aamodt, A., Ernst, G., Storm, J.
592 F., & Juel, B. E. (2025). A repeated awakening study exploring the capac-
593 ity of complexity measures to capture dreaming during propofol sedation.
594 *Scientific Reports*, 15, 32012.

595 Xie, L., Kang, H., Xu, Q., Chen, M. J., Liao, Y., Thiyagarajan, M., ... &
596 Nedergaard, M. (2013). Sleep drives metabolite clearance from the adult
597 brain. *Science*, 342(6156), 373-377.

598 Norton, J. D. (2005). Eaters of the lotus: Landauer's principle and the return
599 of Maxwell's demon. *Studies in History and Philosophy of Science Part B:*
600 *Studies in History and Philosophy of Modern Physics*, 36(2), 375-411.

601 Landauer, R. (1961). Irreversibility and heat generation in the computing
602 process. *IBM Journal of Research and Development*, 5(3), 183-191.

603 Kibble, T. W. B. (1976). Topology of cosmic domains and strings. *Journal*
604 *of Physics A: Mathematical and General*, 9(8), 1387-1398.

605 Sakaguchi, H. (1988). Cooperative phenomena in coupled oscillator systems
606 under external fields. *Progress of Theoretical Physics*, 79(1), 39-46.

607 Still, S., Sivak, D. A., Bell, A. J., & Crooks, G. E. (2012). Thermodynamics
608 of prediction. *Physical Review Letters*, 109(12), 120604.

609 Kawai, R., Parrondo, J. M. R., & Van den Broeck, C. (2007). Dissipation:
610 The phase-space perspective. *Physical Review Letters*, 98(8), 080602.

611 Banach, S. (1922). Sur les opérations dans les ensembles abstraits et leur ap-
612 plication aux équations intégrales. *Fundamenta Mathematicae*, 3(1), 133-
613 181.

- 614 Shenker, O. R. (2000). Logic and entropy. *Philosophy of Science*, 67, S146-
615 S152.
- 616 Sznitman, A.-S. (1991). Topics in propagation of chaos. In *École d'Été de*
617 *Probabilités de Saint-Flour XIX—1989*, Lecture Notes in Mathematics,
618 vol. 1464, pp. 165–251. Springer. doi:10.1007/BFb0085169.